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Key Points:

- A six-level framework of artificial intelligence agent autonomy for hydrologic modeling is defined, with corresponding roles and functions
- A Level-4 hydrologic modeling agent was developed to self-configure simulations, analyze results, and generate reports upon user requests
- Application of near-instant simulation of the July 2025 Texas flood shows current warn-on-forecast limitations compared to observations

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

M. Chen and Y. Hong,
mchen15@ou.edu;
yanghong@ou.edu

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Author Contributions:

Conceptualization: Songkun Yan, Mengye Chen, Zhi Li, Yixin Wen, Mofan Zhang, Jie Cao, Yang Hong
Data curation: Songkun Yan, Mengye Chen, Zhi Li, Siyu Zhu
Formal analysis: Songkun Yan, Mengye Chen, Xiaodong Chen
Funding acquisition: Yang Hong
Investigation: Songkun Yan, Zhi Li, Yixin Wen, Mofan Zhang
Methodology: Songkun Yan, Mengye Chen, Zhi Li, Yixin Wen, Siyu Zhu, Mofan Zhang, Jie Cao
Project administration: Yang Hong

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AI Agent for Hydrologic Modeling: Definition, Development, and Application

Songkun Yan¹ , Mengye Chen¹ , Zhi Li^{2,3} , Yixin Wen⁴ , Siyu Zhu¹, Mofan Zhang⁵ , Di Liu⁶, Jie Cao⁷ , Xiaodong Chen^{1,8} , Chengbin Deng^{6,9} , Tiantian Yang¹ , and Yang Hong¹ 

¹School of Civil Engineering and Environmental Science, University of Oklahoma, Norman, OK, USA, ²Department of Civil, Environmental, and Architectural Engineering, University of Colorado Boulder, Boulder, CO, USA, ³Institute of Arctic and Alpine Research, University of Colorado Boulder, Boulder, CO, USA, ⁴Department of Geography, University of Florida, Gainesville, FL, USA, ⁵Civil and Environmental Engineering, Stanford University, Stanford, CA, USA, ⁶Center for Spatial Analysis, University of Oklahoma, Norman, OK, USA, ⁷School of Computer Science, University of Oklahoma, Norman, OK, USA, ⁸School of Meteorology, University of Oklahoma, Norman, OK, USA, ⁹Department of Geography and Environmental Sustainability, University of Oklahoma, Norman, OK, USA

Abstract Hydrologic modeling supports flood forecasting and water resources management, but complex preprocessing, parameterization, and configuration limit broader use. This study defines a six-level framework for artificial intelligence (AI)-agent autonomy in hydrologic modeling and develops a Level-4 agent, powered by large language models, that translates natural-language requests into data retrieval, model execution, diagnostics, and reports with human oversight. In a proof-of-concept application to the July 2025 Texas flash flood, the agent reproduced key flood dynamics in the tested basin and reduced manual workflow effort. Observation-driven simulations aligned with streamflow records, whereas a forecast-driven run missed the flood response because the deterministic rainfall forecast displaced the storm core. These results suggest the feasibility of AI-agent-assisted hydrologic modeling in the tested case, while robustness and generalization across broader basins and events remain to be established through systematic validation, probabilistic meteorological forcing, and expert review of automated outputs.

Plain Language Summary Hydrology models help scientists and forecasters understand and predict flooding, but running a model usually requires specialized training, complex data preparation, and technical knowledge that limits their use outside expert circles. This study shows how artificial intelligence (AI) agents, powered by large language models, can make flood modeling more accessible. This work designs an AI “hydrology assistant” that accepts plain-language requests such as, “Can you analyze the earlier July 2025 flood in Kerr County, Texas?” The agent automatically retrieves rainfall data, sets up the model, runs simulations, and produces inundation maps, graphs, and reports. In the tested Texas case, the agent completed the end-to-end modeling workflow within minutes and reproduced the main timing and shape of the observed flood hydrograph. The findings suggest that AI agents may help reduce the technical effort required to run flood models, while broader testing is still needed to evaluate robustness across different basins and events. With improved probabilistic weather forecasts and continued human oversight, similar systems may eventually support real-time flood analysis.

1. Introduction

Hydrologic modeling has long been a cornerstone of flood forecasting and water management (Singh & Woolhiser, 2002). Over the past half-century, numerous models have been developed, each with unique strengths, complexities, and limitations (Ruddell & Wagener, 2015). Yet their adoption in both education and operations remains limited by steep learning curves. Running hydrologic models typically requires weeks to months of training, extensive data, parameterization expertise, and proficiency in programming languages and uncommon interfaces (Martel et al., 2017). In classrooms, educators often simplify by using conceptual models or synthetic data (Aghakouchak & Habib, 2010). These simplifications reduce realism, and thus, distancing students and the public from real-world applications. Operational forecasters face similar barriers: National Weather Service staff must configure and run models in a timely manner, yet steep technical requirements and lengthy setup times can hinder their ability to provide rapid, data-driven forecasts during high-impact events, such as extreme rainfall resulted flashfloods. What if we could maintain the complexity and accuracy of hydrologic modeling while

Resources: Songkun Yan, Zhi Li, Yixin Wen, Yang Hong
Software: Songkun Yan, Zhi Li
Supervision: Mengye Chen, Zhi Li, Jie Cao, Xiaodong Chen, Chengbin Deng, Yang Hong
Validation: Songkun Yan
Visualization: Songkun Yan
Writing – original draft: Songkun Yan, Mengye Chen
Writing – review & editing: Songkun Yan, Mengye Chen, Zhi Li, Yixin Wen, Siyu Zhu, Mofan Zhang, Di Liu, Jie Cao, Xiaodong Chen, Chengbin Deng, Tiantian Yang, Yang Hong

reducing its barrier to learning and operating it? The recent advances of the artificial intelligence (AI) agent provide an option.

Recent rapid advances in large language models (LLMs) and research toward artificial general intelligence offer a way to preserve scientific rigor while lowering barriers to use (Bhattacharya et al., 2024; Kalyan, 2024; Minaee et al., 2025; Yan et al., 2023; Yin et al., 2024). LLMs excel at natural language understanding, information integration, and orchestration of complex tasks, which enables the design of AI agents for scientific applications (Kim et al., 2025; Remadi et al., 2024; Tran et al., 2025). In recent years, researchers have proposed multiple agent architectures (ranging from single-agent planners to multi-agent systems) to translate high-level instructions into orchestrated workflows for scientific and engineering tasks. Surveys of autonomous and scientific agents increasingly emphasize modularity, memory, tool use, and domain grounding (Ren et al., 2025; L. Wang et al., 2024).

In the specific domains of hydrology and disaster management, pioneering efforts have begun to leverage these AI capabilities. For instance, HydroPML (Q. Xu et al., 2024) established a unified platform for physics-aware machine learning, effectively bridging data-driven methods with process-based hydrological paradigms. On the application side, DisasterResponseGPT (Goecks & Waytowich, 2023) demonstrates the power of LLMs in rapidly generating text-based action plans for emergency response, while DisasterM3 (J. Wang et al., 2025) utilizes vision-language models to enable global-scale multi-hazard assessment and reasoning from satellite imagery. While these studies have significantly advanced specific components, such as physics-aware modeling, visual perception (J. Wang et al., 2026), and qualitative planning, there remains a critical gap in developing autonomous agents capable of orchestrating the end-to-end quantitative hydrologic modeling workflow.

This gap underscores the urgent need to explore and formalize how AI agents can support domain-specific workflows, moving beyond individual tasks to full-process automation (Sun et al., 2024; Yan et al., 2025). Given the well-recognized challenges in hydrologic modeling complexity, integrating such an agent offers a promising pathway to democratize access for the general public (Ding et al., 2025; Shi et al., 2025). Addressing this need, this study introduces a framework of six autonomy levels for hydrologic modeling agents and develops a Level-4 agent prototype. The system is applied to the July 2025 Texas flash flood as a proof-of-concept demonstration, primarily to evaluate the technical feasibility of LLM-driven workflow orchestration and its potential to lower barriers for research and education.

2. Methods and Materials

2.1. Definition of Six Levels of AI Agent

Agent design has evolved from rule-based controllers to learning-based agents and, most recently, LLM-driven agents (Huang, 2024; Petrović, 2018). Among them, classification schemes commonly use dimensions like agent autonomy and user involvement to distinguish levels of responsibility and decision authority (Feng et al., 2025). In this work, we adopt a two-dimensional framework (Figure 1, from L0 to L5) to guide our agent design for hydrologic modeling, balancing automation with human oversight. In detail, L0 performs no automation; L1 executes user-defined steps; L2 assembles the workflow while the user reviews checkpoints; L3 recommends data sets, parameters, and diagnostics; L4 acts as a task delegate—given a short event description, it infers the basin, fetches forcings, runs, visualizes, and seeks approval; and L5 provides closed-loop autonomy, with continuous monitoring, model updates, run execution, and auditable alert/decision outputs, entirely without human intervention.

With current LLM capabilities, developing an L4 hydrologic modeling agent is realistic and promising, though it presents distinct trade-offs compared to traditional methods. While L4 democratizes access and automates routine “data engineering” tasks (addressing the inefficiency of L0), it introduces the risk of probabilistic errors or “hallucinations” inherent to LLMs. Therefore, under the Level-4 framework, the human role shifts from an “execution-focused operator” to a “validation-focused supervisor.” This “User as Approver” mechanism ensures that while the workflow is streamlined, the final accuracy and physical plausibility are rigorously maintained by human experts. By contrast, fully autonomous L5 remains difficult due to these reliability constraints, as high-stakes actions (e.g., flash-flood warnings) continue to require non-negotiable human accountability.

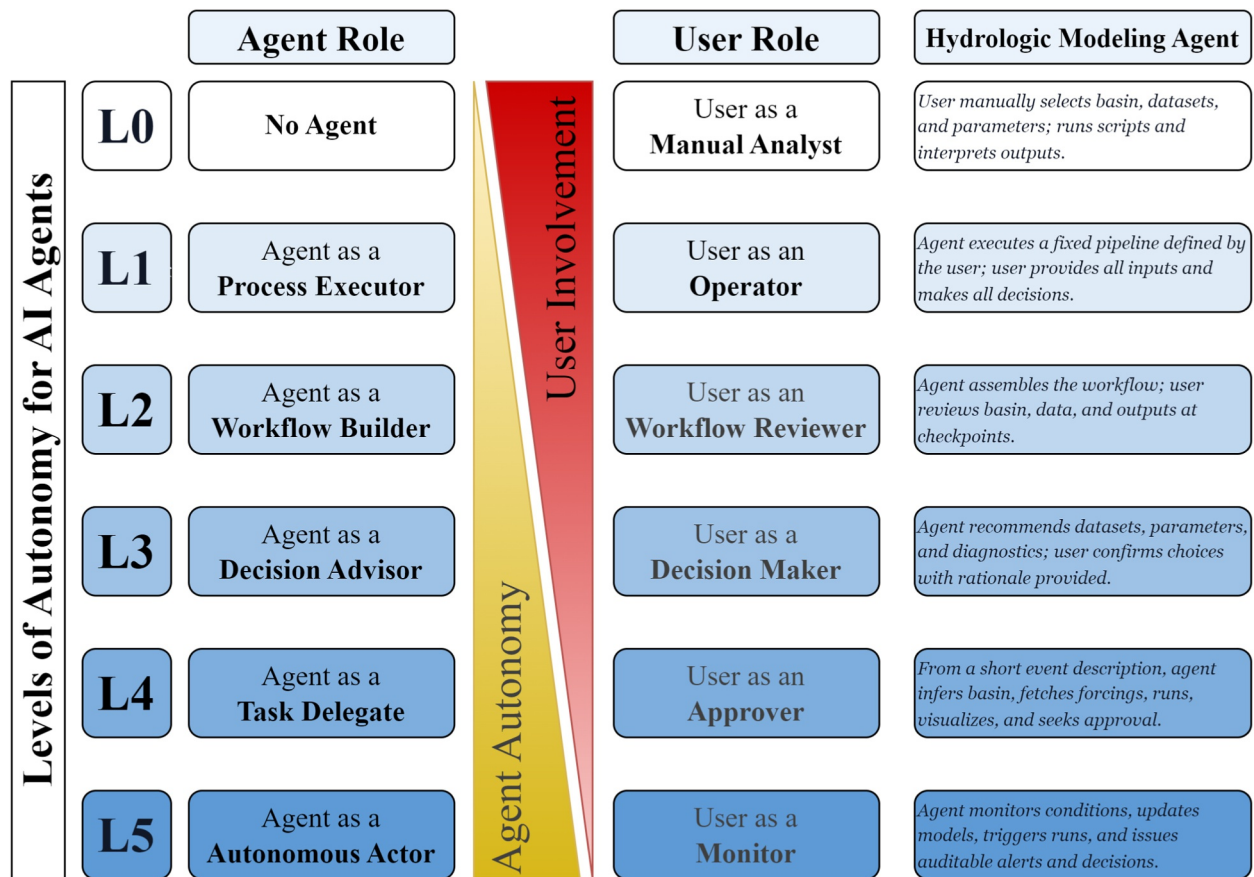


Figure 1. Classification of artificial intelligence agents for hydrologic modeling by agent autonomy and user involvement (L0–L5). The left column lists the agent role; the middle column shows the corresponding user role; and the right column illustrates typical hydrologic-model tasks at each level, from no agent/manual analyst to an autonomous actor. The central wedge denotes decreasing user involvement (red) and increasing agent autonomy (yellow) with higher levels.

2.2. Level-4 Hydrologic Modeling AI Agent System Architecture

Building on the L0–L5 framework in Figure 1, we instantiate an L4 hydrologic modeling AI agent that acts as a task delegate. Figure 2 is a schematic diagram of the L4 AI agent for hydrologic modeling. Figure 2a sketches the end-to-end pipeline from a plain-language query to data retrieval and preprocessing, model configuration and run, visualization/diagnostics, and report generation. Figure 2b shows the multi-agent orchestration, in which an orchestrator coordinates agents for context parsing, data access, outlet and parameter configuration, run control and diagnostics, reporting, and feedback, with auditable data and control flows.

The first AI agent is the Context Parser Agent (A_{CP}), which ingests free-form user queries in any language and converts them into structured model metadata (e.g., geographic extent, time window, event type). The A_{CP} is implemented with LLM APIs (e.g., GPT-4o), which provide robust parsing, entity and time normalization, and tolerance to typos or ungrammatical text. For example, the user types “There is a flash flood happening in central Texas earlier July 2025. Could you help me analyze the flood?” The A_{CP} parses the request, optionally verifies the context against news sources, and extracts and normalizes the metadata to “Kerr County, TX,” “03 July 2025–10 July 2025” and returns an event-relevant anchor coordinate via LLM-guided search for subsequent basin retrieval.

Data set Retriever Agent (A_{DR}) and its related process automation are the core of the system that replaces the heavy manual pre-processing with AI-supported automation. A_{DR} receives the information from A_{CP} and then downloads the data (Section 2.3) that intersects the metadata, for example, the DEM over Kerr County, TX, and climate data from 03 July 2025–10 July 2025. A few pre-processing functions are incorporated, including unit conversion (e.g., K to °C, ft³/s to m³/s), format and numeric-type conversion (e.g., NetCDF to GeoTIFF; int8 to float32), extraction of basin boundary, geo-data clipping, and visualization. All processed data sets are saved for

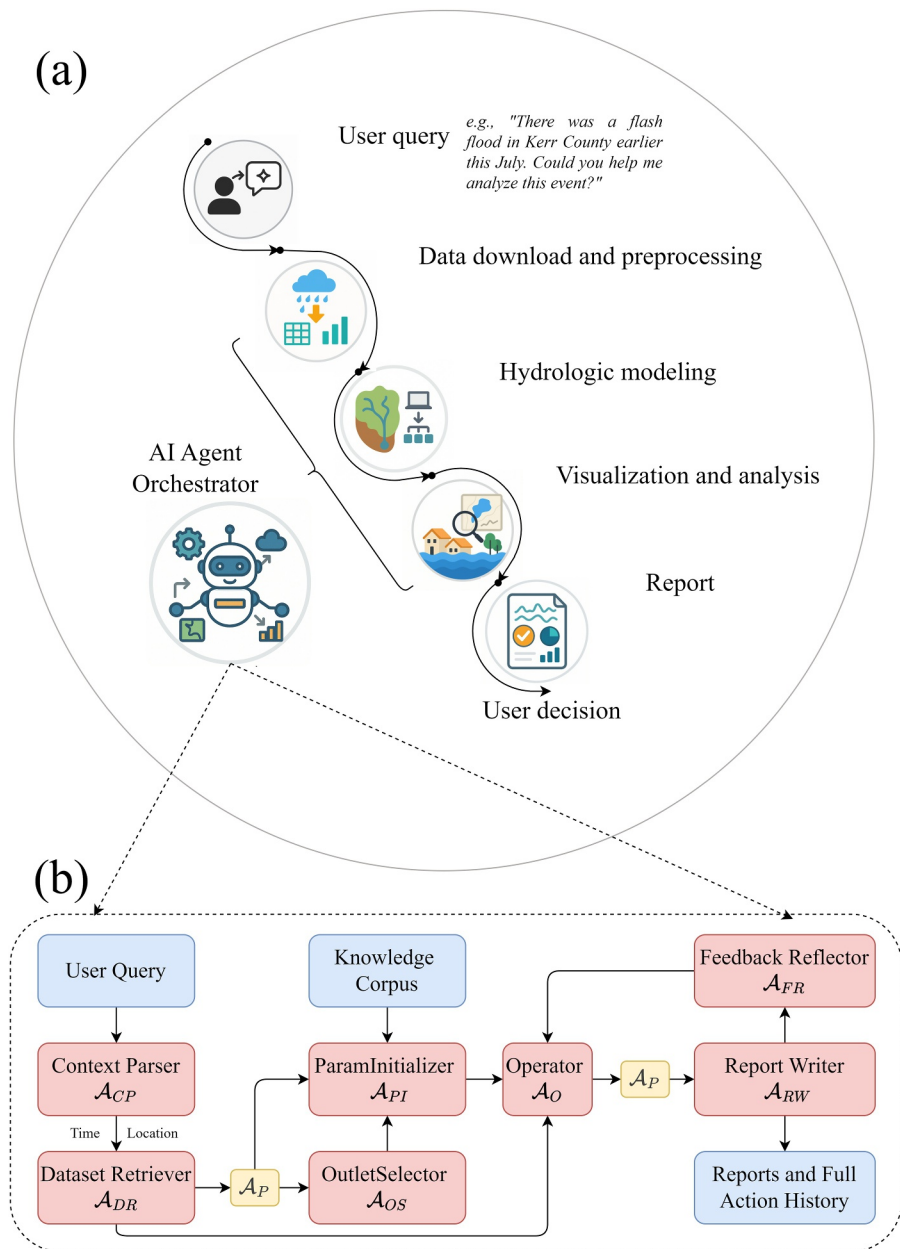


Figure 2. L4 hydrologic-modeling agent. (a) End-to-end pipeline: a plain-language query is translated into data retrieval and preprocessing, model execution, visualization/analysis, and report generation, with the user approving key checkpoints. (b) Multi-agent orchestration showing modules for context parsing, data access, configuration (e.g., outlet and parameters), operation/diagnostics, reporting, and feedback, with auditable flows between components.

subsequent simulation with provenance recorded, and the visualizations are shown to the user to confirm the automatically selected focus region and are forwarded to the Perception Agent ($\mathcal{A}_P$) for downstream interpretation.

Perception Agent ($\mathcal{A}_P$), combining with OutletSelector Agent ($\mathcal{A}_{OS}$), and ParamInitializer Agent ($\mathcal{A}_{PI}$), solves the following two pre-processing tasks for model simulations: selecting the outlet of the focus region/basin, and setting the initial model parameters. To achieve these tasks, the $\mathcal{A}_P$ first reads the map generated by $\mathcal{A}_{DR}$ using a vision-augmented LLM to extract quantitative descriptors of basin morphology, drainage structure, and candidate outlets (USGS stream gauges for the USA). $\mathcal{A}_{OS}$ buffers the target basin and retrieves all stream gauges located within or adjacent to the polygon. It then uses the system's visualizations (DEM, flow-accumulation with gauges)

together with prompt-encoded hydrologic rules (predefined in the agent system) to recommend the most plausible outlet gauge and presents it for user confirmation. Using retrieval-augmented generation, A_{PI} prioritizes extracting parameters from the CONUS-wide calibrated data set established by a previous Coupled Routing and Excess Storage (CREST) calibration study (Chen et al., 2023), which covers over 3,000 USGS basins. For basins not explicitly included in this library, the agent employs a physically grounded regionalization strategy, deriving parameters from spatially proximal or hydrologically similar basins rather than relying solely on generalized LLM priors. Furthermore, to ensure the physical validity of initial conditions, the agent autonomously retrieves meteorological forcing for a designated warmup period (e.g., 1 month) prior to the simulation start date. This process generates equilibrated model states (e.g., soil moisture), thereby minimizing the uncertainty associated with arbitrary cold starts.

Operator Agent (A_O) aggregates outputs from upstream agents and emits the model control file that governs execution (run options, scheduling, I/O routing). It orchestrates runs, computes post-simulation performance statistics using standard error and skill metrics, and conducts preliminary hydrologic assessments, including flood return period, Flashiness Intensity, and Flashiness–Intensity–Duration–Frequency (F-IDF, Li et al., 2023).

Report Writer Agent (A_{RW}) compiles visualizations, simulation outputs, and diagnostics into a structured report, including hydrograph interpretation and key findings. The narrative follows prompt-encoded templates and rubrics, so depth is bounded by the provided context and may not include expert-level analysis beyond the inputs. With the support of the Feedback Reflector Agent (A_{FR}) that parses the user's plain language comments and feedback, the internal loop can improve the report writing by including human corrections or improve the model simulation by giving expert insights into the model setting processes.

The multi-agent architecture turns plain-language requests into an end-to-end hydrologic modeling workflow, requiring minimal prior expertise. From a single query, the system identifies the target basin, acquires and preprocesses data, runs the model, and delivers basin visualizations, precipitation maps and animations, 2D streamflow animations, inundation maps, hydrographs, skill metrics with hydrologic analyses, and a structured report for user approval. The current AI agent system can achieve the L4 AI agent level described in Figure 1.

2.3. Hydrologic Model and Data

The CREST model was developed to enhance hydrologic sciences and applications at high spatiotemporal scales, a significant advancement in assimilating multi-source data into hydrologic models (Chen et al., 2020, 2021; Clark et al., 2017; Flamig et al., 2020; Li, Chen, et al., 2022; J. Wang et al., 2011; Wen et al., 2021). It facilitates quasi-global flood modeling by integrating remote-sensing data, marking a pioneering operational deployment in real-time flood prediction across continents with the National Weather Service (Gourley et al., 2017). Within the CREST model family, CREST version 3.0 integrates a conceptual groundwater module, enabling comprehensive hydrologic cycle simulations, crucial for accurate streamflow modeling and water storage dynamics understanding, thereby ensuring a representation of surface water-groundwater interactions. The model underwent an extensive CONUS-wide calibration process, notably improving accuracy in the eastern CONUS (Chen et al., 2023; Shen et al., 2017; Vergara et al., 2016). CREST-iMAP is a 2D hydrological and hydraulic coupled model and is a branch member of the CREST modeling family (Chen et al., 2021, 2022; Li et al., 2021; Li, Gao, et al., 2022). The model can produce channel discharge, inundation extent, water flux, flow velocity, and flood depth.

To ensure data reliability and minimize hallucination risks, the AI agent employs a “Selection by Authority” strategy. This restricts automated retrieval to data sets characterized by (a) institutional provenance (e.g., official agencies like USGS and NASA), (b) programmatic stability via official APIs or RESTful web services, and (c) extensive peer-reviewed validation. Following these criteria, all data sets used in this study are openly available in the United States and can be accessed programmatically via APIs, enabling our AI agent to automatically retrieve them for hydrologic applications. Specifically, precipitation data are obtained from the Multi-Radar-Multi-Sensor (MRMS) 1-hr gauge-corrected Quantitative Precipitation Estimates (QPEs) (Qi et al., 2016; Smith et al., 2016; Zhang et al., 2016), archived at the Iowa State University Environmental Mesonet (IEM). Stream gauge data and watershed attributes are derived from the GAGES-II database (Geospatial Attributes of Gages for Evaluating Streamflow), maintained by the U.S. Geological Survey (USGS). In addition, time series of streamflow observations (6,036 stations) are obtained via the USGS Representational State Transfer (REST) web service using the Instantaneous Values (IV) Python package. Satellite-based precipitation estimates are incorporated from the

Integrated Multi-satellite Retrievals for GPM (IMERG) data set (Huffman et al., 2015, 2020; Zhu et al., 2024), acquired from NASA's Goddard Earth Sciences Data and Information Services Center (GES DISC). River network information is extracted from HydroRIVERS v1.0 for North America, while topographic and hydrologic conditioning layers, including the digital elevation model, flow direction, and flow accumulation, are provided by the HydroSHEDS core data sets. The reliability and utility of these data sets for distributed hydrologic modeling and benchmarking have been extensively validated in prior research (Flamig et al., 2020; Lehner et al., 2008; Morsy et al., 2018; Newman et al., 2015; Seo et al., 2021; Velásquez et al., 2025; J. Xu et al., 2022).

This study contains no human subjects, human data or tissue, or animals.

3. Application

3.1. July 2025 Central Texas Flash Floods

This Level-4 AI agent is applied to the flash flood in early July 2025 in Kerr County (Figure 3a), Texas, which severely impacted communities along the upper Guadalupe River, including the Camp Mystic area. This event was widely covered in local and national media due to its rapid onset and damaging consequences. By typing “There was a flash flood in Kerr County earlier July 2025. Could you help me analyze this event?,” the system initiated its reasoning workflow. Crucially, the agent employed visual-spatial reasoning to refine the simulation domain rather than relying solely on keyword matching. Upon initialization, it evaluated the spatial coverage of the identified basin against the user's broader “county-wide” request. Determining that the initial boundary was insufficient, the agent autonomously “expanded” the domain to a larger contributing area to capture the full hydrological context. Following this confirmation, it retrieved, clipped, and visualized the MRMS QPEs for the event, providing users with a spatiotemporal view of the rainfall pattern (Figure 3e).

3.2. AI Agent Simulation Results and Analysis

Figures 3a–3c presents the agent generated visualizations for the July 2025 event. The system identifies the Kerr County basin and displays the river network, USGS gauges, Camp Mystic, and the DEM, highlighting the southern branch as the dominant contributor during the storm. The simulated inundation from CREST-iMAP in Figure 3c concentrates along the Guadalupe main stem near Camp Mystic and aligns with the mapped low-lying floodplain. Figure 3d summarizes the temporal evolution. The basin-integrated precipitation volume rises rapidly to about 2.2×10^7 m³/hr near 09:00 UTC and then declines, while the simulated discharge at gauge 08165500 peaks near 2,400 m³/s around 10:00 UTC, indicating a response time of roughly 1 hr for the 744.5 km² drainage area. The peak precipitation during this event concentrated on the south branch of the basin. Between 4:00 and 20:00 UTC (03 July 2025 11:00 p.m. to 04 July 2025 3:00 p.m. CDT), 0.17 km³ of rain, about one-third of the 2024–2025 hydrological-year total, fell on the upper Guadalupe River and caused the devastating flood.

Figure 3e depicts the spatiotemporal evolution of the event: MRMS precipitation intensifies over the southern branch and the CREST simulation activates the mainstem and tributaries; in the agent workflow this panel is delivered as a synchronized animation. Figure 4b is also a standard agent output, presenting hydrographs at USGS 08165300 and 08165500. Using AI-suggested initial parameters (informed by a prior calibration data set, model documentation, and local hydro-geophysical context), the simulation reproduces the observed flood with relatively strong skill (NSE = 0.86, CC = 0.93 at 08165300 and NSE = 0.74, CC = 0.98 at 08165500), capturing the sharp rise and recession consistent with thin soils and limited storage.

The flashiness index of 16.3 mm/hr² was automatically calculated for the case study. Compared with the historical USGS gauge (08165500) record, the July 2025 flood event was indeed unprecedented, which was faster and larger than any flood event in this basin. The FIDF (Li et al., 2023) analysis gave this flood event an “infinite” return period, considering the “normal” streamflow for this gauge is around 0.25–0.5 m³/s, and suddenly rose to 2,421.1 m³/s in 4 hr on 4 July 2025. The AI-generated report stated that “The flood was triggered by short, intense storms on 4 July, concentrated directly over the watershed,” “The observed peak discharge was record-breaking and not represented in historical return period data sets,” and “Hydrologic simulation reproduced the timing and shape of the flood hydrograph, with good overall agreement, though the peak magnitude did not fully match the observed magnitude. This highlights the importance of accurate rainfall representation and continued model optimization in flash-flood-prone basins.” The first two opinions summarize the understanding of the hydrological response well, whereas the latter point, made without supporting tests or evidence, reflects a limitation in

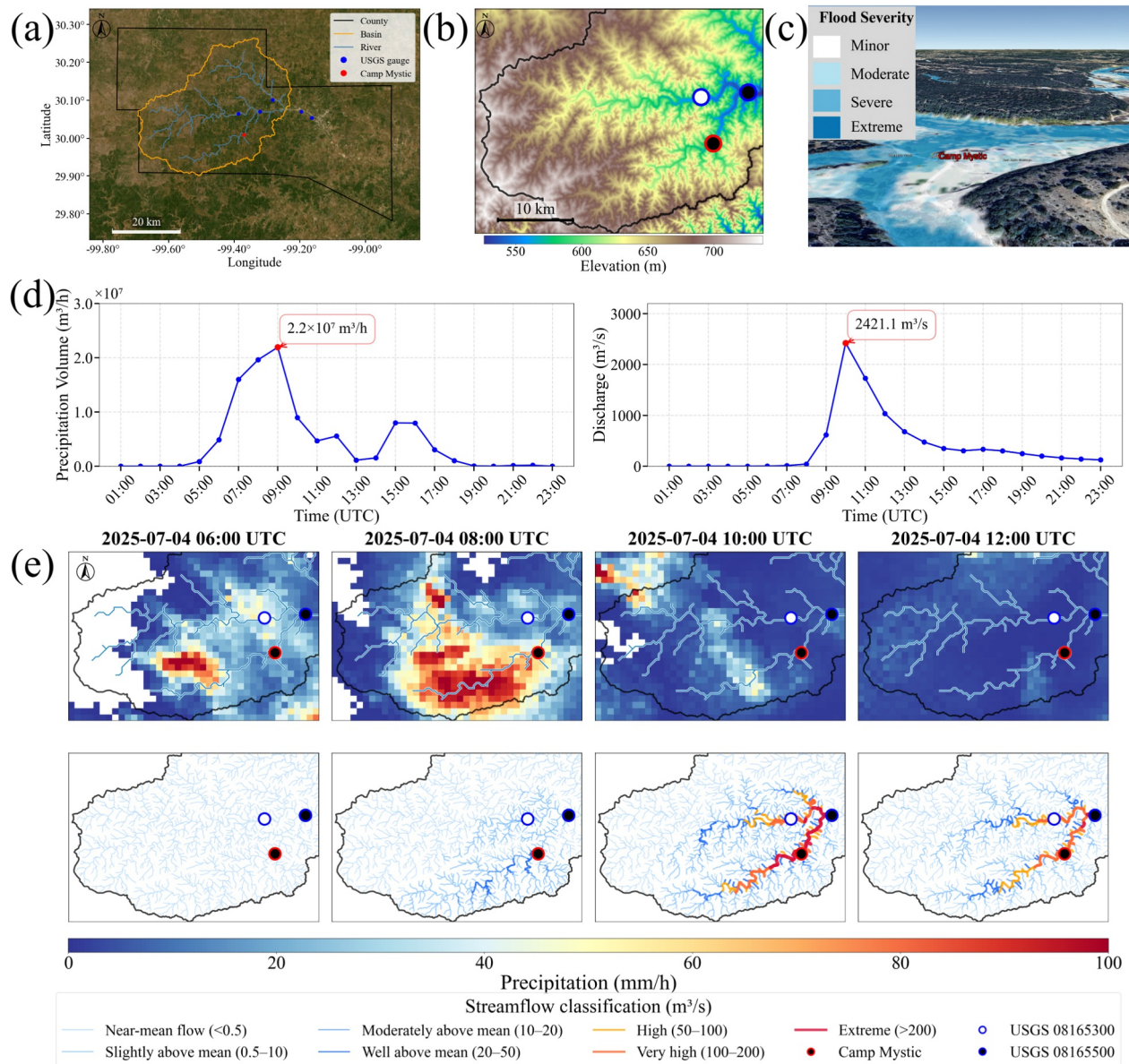


Figure 3. 4 July 2025 central Texas flash flood. (a, b) Study area. Basin location, rivers, gauges, Camp Mystic, and DEM. (c) Illustrative flood inundation visualization near Camp Mystic, simulated by CREST-iMAP and rendered via Google Earth. (d) Time series of basin-integrated precipitation volume (left) and simulation discharge at USGS 08165500 (right, outlet). (e) Snapshots at 06:00, 08:00, 10:00, and 12:00 UTC; top shows Multi-Radar-Multi-Sensor precipitation, bottom shows simulated streamflow magnitude.

the agent's analytical depth. This suggests that future development should emphasize more rigorous Quantitative Precipitation Forecast (QPF) data prediction and enhanced hydrological reasoning by the agent.

4. Discussion

4.1. Strengths and Limitations of Level-4 AI Agent

The proposed Level-4 AI agent introduces a distinct paradigm shift in hydrologic modeling, offering advantages primarily in accessibility and research efficiency rather than mere execution speed. As summarized in Table 1, the agent significantly lowers the technical barrier by abstracting complex data retrieval and model configuration tasks, which traditionally require proficiency in Python or Linux, into a natural language interface. These quantitative entries are intended to characterize the typical successful Kerr County workflow; failed runs and

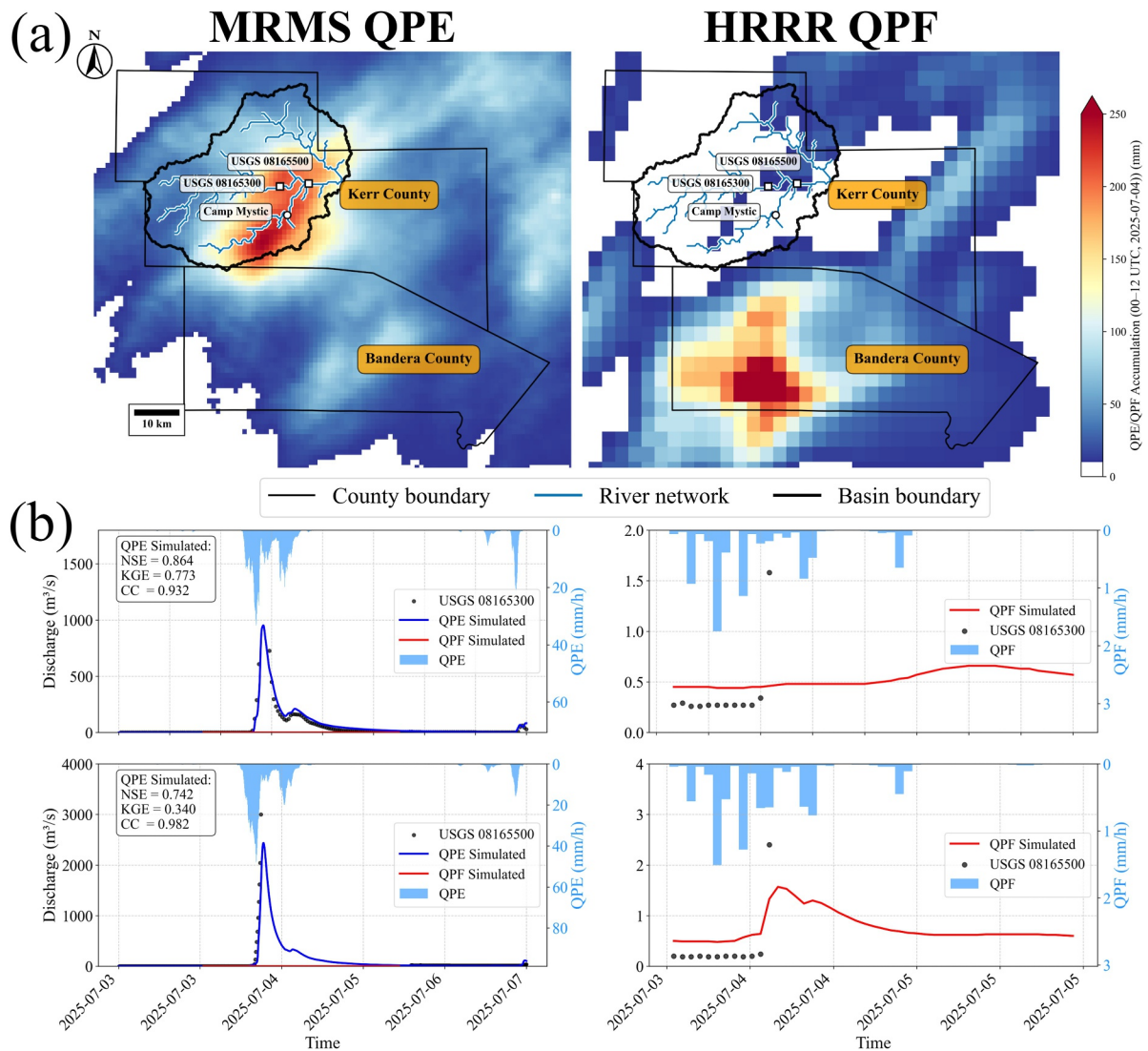


Figure 4. Precipitation inputs and simulated discharge. (a) Event-total Multi-Radar-Multi-Sensor Quantitative Precipitation Estimate (observation-based estimate) and High-Resolution Rapid Refresh Quantitative Precipitation Forecast (model forecast) over the study basin; county/basin boundaries, river network, Camp Mystic, and USGS gauges 08165300 and 08165500 are indicated. (b) Coupled Routing and Excess Storage simulations at both gauges, distinguishing between observation-driven results (Blue line, “QPE Simulated”) and forecast-driven results (Red line, “QPF Simulated”); bars show the corresponding precipitation time series and dots show USGS observations. The figure distinguishes estimation-based versus forecast-based forcings and their paired hydrographs.

corrective re-prompts are discussed separately in the reproducibility assessment below. This democratizes access to advanced tools, allowing individuals with limited programming expertise, such as emergency managers or the general public, to analyze hydrologic processes and potentially improving public awareness. Furthermore, the agent provides preliminary evidence of adaptability through its “zero-shot” configuration capability. Unlike rigid scripts that often require manual refactoring for new locations, the agent can dynamically infer spatiotemporal parameters from vague queries and adapt data retrieval and modeling logic for selected test events (Table S1 in Supporting Information S1). However, these zero-shot tests should be interpreted as an initial feasibility assessment rather than conclusive evidence of robustness or generalization across all basins, flood mechanisms, or data environments.

Despite these operational strengths, the reliability of the agent remains contingent upon the quality of input data and the intrinsic probabilistic nature of LLMs. Atypical hydrological conditions or data irregularities can occasionally introduce uncertainty into the agent’s reasoning chain. Therefore, the system is designed to automate the workflow execution but not to replace hydrologic judgment. To evaluate workflow reproducibility, we

Table 1
Comparison of Operational Paradigms Between the Traditional Script-Based Workflow and the Proposed Artificial Intelligence Agent Workflow

Comparison dimension	Traditional approach (manual/scripting)	AI agent approach (proposed)	Core value proposition
Prerequisite expertise	High: Demands dual proficiency in hydrology and advanced programming (e.g., Python, Linux shell) for data processing and model configuration	Low: Requires primarily hydrological domain knowledge; technical complexities are abstracted via a Natural Language Interface	Accessibility: Democratizes advanced modeling tools by lowering the technical barrier to entry for non-coders
Adaptability to new events	Rigid: Workflows often rely on hard-coded parameters; applying to new locations requires manual code refactoring and data source reconfiguration	Flexible: Autonomously infers spatiotemporal parameters from natural language; adapts data retrieval and modeling logic dynamically (zero-shot)	Rapid response: Enables immediate deployment to new basins without manual code modification
Human responsibility	Operator (Execution-focused): Human effort is consumed by data formatting, debugging scripts, and managing file I/O	Supervisor (validation-focused): Human effort is shifted to high-level reasoning, validating logic, and interpreting hydrological insights	Research efficiency: Reduces operational toil, allowing researchers to focus on scientific synthesis rather than data engineering
Manual interventions	High (20+ discrete steps): Requires continuous manual execution of scripts, API requests, data format conversions, and parameter tuning	Low (1–3 steps): Requires only initial natural language prompts and human-in-the-loop approval at key checkpoints	Operational automation: Drastically reduces manual operational overhead and the risk of human error during setup
Estimated time consumption	<2 Working days: Data retrieval, preprocessing, cold-start equilibration, and debugging consume the majority of the time	Minutes (~5–10 min): Autonomous end-to-end orchestration seamlessly progresses from query to report generation	Rapid deployment: Enables near real-time hydrologic analysis, critical for rapid disaster response

Note. Quantitative benchmarks are approximate estimates based on the July 2025 Kerr County flash flood case study, rather than rigorous computational performance benchmarks. A “manual intervention” refers to a user action to approve, re-prompt, restart, or manually correct the workflow. The 5–10 min estimate refers to the typical successful run from the finalized natural-language query to report generation under the same workflow environment.

repeated the Kerr County workflow 20 times under the same prompt structure and workflow environment. A first-pass successful run was defined as completion of the end-to-end workflow, including context parsing, data retrieval, model execution, visualization, and report generation, without corrective re-prompting, restart, or low-level debugging. Among the 20 trials, 18 were completed successfully on the first pass, corresponding to a success rate of 90%. The failed cases were mainly associated with temporary data-access issues and incorrect representative outlet-gauge selection (requiring a corrective restart to proceed since it occurred prior to user confirmation). This success rate refers to workflow completion rather than hydrological accuracy, which was evaluated separately using streamflow-based metrics. Because this reproducibility test was conducted for the Kerr County case, the reported success rate should be interpreted as case-specific rather than directly generalizable to all basins or events. The human user serves as the final authority to validate the logic, verify the physical plausibility of the outputs, and ensure robust interpretation, reinforcing that the agent functions as a powerful scientific assistant rather than an autonomous decision-maker. Finally, regarding the hydrological core, this study utilizes the CREST model as a representative testbed. While we acknowledge that dynamic model selection and spatio-temporal hydraulic modeling are valuable for optimizing performance and assessing flood impacts, our primary objective is to demonstrate the feasibility of the AI automation framework itself using the foundational rainfall-runoff process. Although the underlying framework supports hydraulic capabilities (as visually illustrated in Figure 3c), we prioritized validating the agent's reasoning on streamflow simulation. Importantly, the proposed architecture is designed to be model-agnostic and extensible, allowing for the seamless integration of diverse hydrological models and hydraulic modules in future iterations to address varying basin characteristics and user requirements.

Positioned primarily as a methodological proof-of-concept, this study highlights a paradigm shift from manual data engineering to AI-orchestrated modeling. The current validation, including the Kerr County reproducibility test and additional zero-shot event demonstrations, should be interpreted as preliminary evidence of feasibility and workflow adaptability rather than conclusive evidence of generalized robustness. Establishing generalized reliability will require more systematic validation across diverse hydro-climatic regions, basin sizes, flood-generation mechanisms, data availability conditions, and model configurations.

4.2. Forecasting Ability of the AI Agent Limited by QPF

The current AI agent architecture is designed with the flexibility to ingest QPF data sets, such as the High-Resolution Rapid Refresh (HRRR), to drive hydrologic models. This capability demonstrates the framework's potential to transition from retrospective analysis to real-time flood forecasting. To explore this feasibility, we conducted a preliminary test using the operational HRRR forecast available during the July 2025 Texas event.

Figure 4a compares the accumulated precipitation from the reference MRMS QPE against the HRRR QPF. While the agent successfully automated the ingestion of forecast data, the result highlights a critical challenge in hydrologic forecasting: spatial uncertainty. The deterministic HRRR forecast captured the convective intensity but placed the storm core approximately 30 km south of the actual event. Consequently, the hydrologic simulation driven by this single-member forecast (Figure 4b) did not trigger a significant flood peak in the Upper Guadalupe River. This “miss” underscores the limitations of relying on a single deterministic forecast for localized flash floods, where slight spatial displacements can propagate into significant hydrological errors (Chen et al., 2022). Future operational applications would necessitate integrating ensemble-based probabilistic forecasting (e.g., HRRRE) to account for such spatial uncertainties and capture a wider range of potential scenarios. Given these limitations and the critical safety implications of flood forecasting, we emphasize that this “Level-4” agent currently serves as an experimental research demonstration rather than an operational “warn-on-forecast” system. Human experts must remain the final authority to validate all model outputs.

5. Conclusion

This study first defines a six-level framework for AI-agent autonomy in hydrologic modeling, with clear roles and functions that delineate capabilities and governance. Following this framework, a Level-4 agent (user as approver) is developed that configures simulations, analyzes outputs, and generates reports while preserving accountability through human oversight. Application to the July 2025 Texas flash flood (upper Guadalupe basin) demonstrates the Agent reproduces the flood dynamics within minutes and highlights the limitations of HRRR-based warn-on-forecast relative to MRMS observation-driven workflows due to QPF spatial displacement.

Overall, this work provides a proof-of-concept AI-agent framework that can be integrated with existing hydrologic models to lower technical barriers, broaden access to hydrologic modeling, and reduce manual workflow effort in the tested case. Further investigation is needed to evaluate the framework's robustness, scalability, and generalization across broader basins, events, forcing data sets, and levels of autonomy.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The following sources were accessed: MRMS QPE data via the Iowa Environmental Mesonet archive (<https://mtaarchive.geol.iastate.edu/>), USGS GAGES-II watershed attributes (https://water.usgs.gov/lookup/getspatial?gage_sII_Sept2011), USGS streamflow records via the REST Instantaneous Values service (<https://waterservices.usgs.gov/>), GPM IMERG precipitation data via NASA GES DISC (<https://disc.gsfc.nasa.gov/>), HydroRIVERS v1.0 (<https://www.hydrosheds.org/products/hydrorivers>), and HydroSHEDS core data sets (<https://data.hydrosheds.org/>). Code and video demo are available at Zenodo (Yan et al., 2025; <https://doi.org/10.5281/zenodo.20388794>) or GitHub (https://github.com/Skyan1002/AQUAH_v0.3.git).

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